CIE International A Level Physics

# Topic 14, 15 & 16

# Thermal Physics

Revision Booklet

**This booklet covers:**

- Temperature and Thermal Equilibrium
- Temperature Scales
- Specific Heat Capacity and Latent Heat
- The Mole and Ideal Gases
- Kinetic Theory of Gases
- Internal Energy
- The First Law of Thermodynamics

## Thermal Equilibrium

### Thermal Energy Transfer

Thermal (heat) energy is transferred from a region of **higher temperature** to a region of **lower temperature**. Transfer continues until the two regions reach the same temperature.

### Thermal Equilibrium

Two objects are in **thermal equilibrium** when there is no net transfer of thermal energy between them. This occurs when they are at the **same temperature**.

*Zeroth Law of Thermodynamics:* If object A is in thermal equilibrium with object C, and object B is also in thermal equilibrium with object C, then A and B are in thermal equilibrium with each other.

## Temperature Scales

### Thermometric Property

A **thermometric property** is a physical property that varies continuously and measurably with temperature. Examples include:

- Resistance of a metal wire (e.g. platinum resistance thermometer)
- e.m.f. of a thermocouple
- Volume of a gas at constant pressure
- Density of a liquid

### Thermodynamic Temperature Scale

The **thermodynamic (Kelvin) temperature scale** does not depend on the property of any particular substance. It is an absolute scale with:

- **Absolute zero** (0 K) — the lowest possible temperature
- **Triple point of water** (273.16 K) — fixed reference point

### Temperature Conversion

$$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$$

$T$ = thermodynamic temperature (K)
$\theta$ = Celsius temperature (°C)

Note: a temperature *difference* of 1 K equals a difference of 1 °C.

### Absolute Zero

At absolute zero (0 K = −273.15 °C):

- Molecules have minimum possible kinetic energy
- It is impossible to reach in practice, only to approach asymptotically
- All ideal gases would have zero volume (zero pressure at constant volume)

## Specific Heat Capacity and Latent Heat

### Specific Heat Capacity

The **specific heat capacity** $c$ of a substance is the energy required to raise the temperature of **unit mass** by **one kelvin** (or one degree Celsius), without change of state.

$$c = \frac{Q}{m\,\Delta T} \qquad \text{units: } \mathrm{J\,kg^{-1}K^{-1}}$$

### Specific Heat Capacity

$$Q = mc\Delta T$$

$Q$ = thermal energy transferred (J)
$m$ = mass (kg)
$c$ = specific heat capacity ($\mathrm{J\,kg^{-1}\,K^{-1}}$)
$\Delta T$ = temperature change (K or °C)

### Specific Latent Heat

The **specific latent heat** $L$ of a substance is the energy required to change the state of **unit mass** at constant temperature.

$$L = \frac{Q}{m} \qquad \text{units: } \mathrm{J\,kg^{-1}}$$

- **Specific latent heat of fusion** $L_f$: solid → liquid (melting)
- **Specific latent heat of vaporisation** $L_v$: liquid → gas (boiling)

Note: $L_v > L_f$ for any given substance, since greater work is done separating molecules completely during vaporisation.

### Specific Latent Heat

$$Q = mL$$

$Q$ = thermal energy transferred (J)
$m$ = mass (kg)
$L$ = specific latent heat ($\mathrm{J\,kg^{-1}}$)

**Common Mistake**

During a change of state, temperature remains **constant** even though energy is being supplied. The energy goes into breaking intermolecular bonds, not increasing kinetic energy.

## The Mole and Ideal Gases

**Amount of Substance**

The **mole** (mol) is the SI base unit for amount of substance. One mole of any substance contains exactly $N_A = 6.02 \times 10^{23}$ particles (the Avogadro constant).

**Equation of State for an Ideal Gas**

$$pV = nRT \qquad \text{or} \qquad pV = NkT$$

$p$ = pressure (Pa)
$V$ = volume ($m^3$)
$n$ = number of moles (mol)
$R$ = molar gas constant = $8.31\ \mathrm{J\,mol^{-1}K^{-1}}$
$T$ = thermodynamic temperature (K)
$N$ = number of molecules
$k$ = Boltzmann constant = $1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$

The Boltzmann constant: $k = R/N_A$

**Ideal Gas**

An **ideal gas** is one that obeys $pV \propto T$ (where $T$ is thermodynamic temperature) under all conditions. Real gases approximate ideal behaviour at low pressures and high temperatures.

**Gas Laws — special cases of $pV = nRT$**

- **Boyle's Law** (constant $T$): $pV$ = constant, so $p_1V_1 = p_2V_2$
- **Charles' Law** (constant $p$): $V/T$ = constant, so $V_1/T_1 = V_2/T_2$
- **Pressure Law** (constant $V$): $p/T$ = constant, so $p_1/T_1 = p_2/T_2$

**Common Mistake**

Always use **thermodynamic temperature in kelvin** in gas law calculations — never degrees Celsius.

# Kinetic Theory of Gases

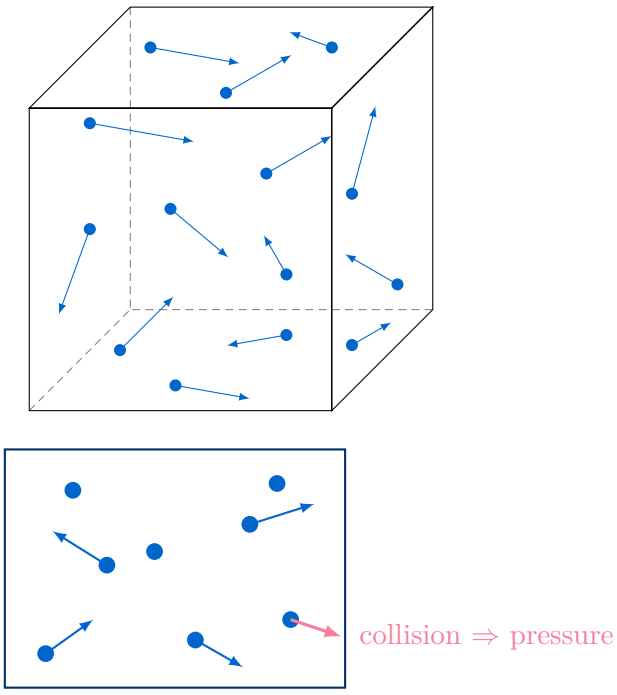
## Assumptions of the Kinetic Model

**Basic Assumptions of Kinetic Theory**

1. The gas contains a **large number** of molecules moving in **random directions** with a range of speeds.
2. The molecules occupy **negligible volume** compared to the volume of the gas.
3. All collisions are **perfectly elastic** (no kinetic energy lost).
4. **Intermolecular forces are negligible** except during collisions.
5. The duration of collisions is **negligible** compared to the time between collisions.
6. Molecules obey **Newton's laws of motion**.

## Pressure from Kinetic Theory

The pressure exerted by a gas arises from molecules colliding with the container walls. Each collision exerts a force; the average of many such collisions gives a steady pressure.




Random molecular motion causes pressure on walls

**Kinetic Theory Equation**

$$pV = \frac{1}{3}Nm\langle c^2\rangle$$

$p$ = pressure (Pa)
$V$ = volume ($m^3$)
$N$ = total number of molecules
$m$ = mass of one molecule (kg)
$\langle c^2 \rangle$ = mean square speed ($m^2\,s^{-2}$)

The **root-mean-square speed**: $c_{\text{r.m.s.}} = \sqrt{\langle c^2 \rangle}$

**Note:** $c_{\text{r.m.s.}}$ is not the same as the mean speed $\bar{c}$. Because we square first then average, faster molecules contribute more. In general $c_{\text{r.m.s.}} > \bar{c}$.

**Derivation of $pV = \frac{1}{3}Nm\langle c^2 \rangle$**

Consider one molecule of mass $m$ moving with speed $c_x$ in the $x$-direction inside a box of side $L$.

**Step 1 — change in momentum at one wall:** The molecule hits the right wall and bounces back elastically:

$$\Delta p = mc_x - (-mc_x) = 2mc_x$$

**Step 2 — time between collisions with the same wall:** The molecule must travel $2L$ before returning:

$$\Delta t = \frac{2L}{c_x}$$

**Step 3 — force from one molecule:**

$$F = \frac{\Delta p}{\Delta t} = \frac{2mc_x}{2L/c_x} = \frac{mc_x^2}{L}$$

**Step 4 — pressure from $N$ molecules:** Summing over all $N$ molecules and dividing by area $L^2$:

$$p = \frac{Nm\langle c_x^2 \rangle}{L^3} = \frac{Nm\langle c_x^2 \rangle}{V}$$

**Step 5 — extend to 3 dimensions:** By symmetry of random motion: $\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle = \frac{1}{3}\langle c^2 \rangle$

$$\boxed{pV = \tfrac{1}{3}Nm\langle c^2 \rangle}$$

## Molecular Kinetic Energy

**Derivation of $\langle E_k \rangle = \frac{3}{2}kT$**

Start with the two equations for an ideal gas:

$$pV = \tfrac{1}{3}Nm\langle c^2 \rangle \qquad \text{and} \qquad pV = NkT$$

Equating the right-hand sides:

$$\tfrac{1}{3}Nm\langle c^2 \rangle = NkT$$

Divide both sides by $N$ and multiply by $\frac{3}{2}$:

$$\tfrac{1}{2}m\langle c^2 \rangle = \tfrac{3}{2}kT$$

Since $\langle E_k \rangle = \frac{1}{2}m\langle c^2 \rangle$:

$$\boxed{\langle E_k \rangle = \frac{3}{2}kT}$$

**Average Translational Kinetic Energy**

$$\langle E_k \rangle = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

The average kinetic energy of a molecule is **directly proportional to thermodynamic temperature**.

**Key Deductions**

- At absolute zero, molecules have minimum kinetic energy
- Molecules of different gases at the same temperature have the **same average KE**
- Heavier molecules move more **slowly** on average at the same temperature: $c_{\text{r.m.s.}} = \sqrt{3kT/m}$
- $c_{\text{r.m.s.}} \propto \sqrt{T}$ — doubling thermodynamic temperature increases r.m.s. speed by factor $\sqrt{2}$

## Internal Energy

**Internal Energy**

The **internal energy** of a system is the sum of the **random kinetic energies** and **potential energies** of all the molecules in the system.

$$U = E_{k,\text{total}} + E_{p,\text{total}}$$

- Internal energy is a **function of state**.
- For an **ideal gas**: $E_p = 0$, so $U = E_{k,\text{total}}$ only.
- A rise in temperature $\Rightarrow$ increase in average KE $\Rightarrow$ increase in $U$.
- During a change of state: temperature (and KE) constant; $E_p$ increases as bonds break; $U$ increases.

## The First Law of Thermodynamics

**Work Done by/on a Gas**

When a gas changes volume at **constant pressure**:

$$W = p\,\Delta V$$

$W$ = work done **by** the gas (J)
$p$ = pressure (Pa)
$\Delta V$ = change in volume ($m^3$)

### First Law of Thermodynamics

$$\Delta U = q + W$$

$\Delta U$ = increase in internal energy of the system (J)
$q$ = energy supplied **to** the system by heating (J)
$W$ = work done **on** the system (J)

### Sign Convention

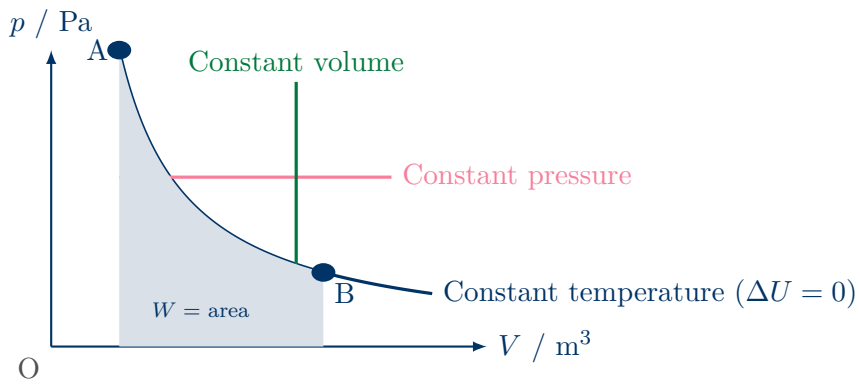
CIE uses $\Delta U = q + W$ where $W$ is work done **on** the system. Some textbooks use $\Delta U = q - W$ where $W$ is work done **by** the system. Always check which convention is being used.

### Special Cases of the First Law

| Process | Condition | Result |
|---|---|---|
| Constant temperature | $\Delta T = 0 \Rightarrow \Delta U = 0$ | $q = -W$ |
| No heat transfer | $q = 0$ | $\Delta U = W$ |
| Constant volume | $\Delta V = 0 \Rightarrow W = 0$ | $\Delta U = q$ |
| Constant pressure | $p$ constant | $\Delta U = q - p\Delta V$ |

## $p$–$V$ Diagrams

The **area under a $p$–$V$ curve** equals the work done by the gas during that process.



## Formula Summary

| Formula | Quantity | Units |
|---|---|---|
| $T/\mathrm{K} = \theta/°\mathrm{C} + 273.15$ | Temperature conversion | K |
| $Q = mc\Delta T$ | Sensible heat | J |
| $Q = mL$ | Latent heat | J |
| $pV = nRT$ | Ideal gas (moles) | Pa, m$^3$, K |
| $pV = NkT$ | Ideal gas (molecules) | Pa, m$^3$, K |
| $k = R/N_A$ | Boltzmann constant | J K$^{-1}$ |
| $pV = \frac{1}{3}Nm\langle c^2\rangle$ | Kinetic theory | Pa |
| $\langle E_k\rangle = \frac{3}{2}kT$ | Mean molecular KE | J |
| $W = p\Delta V$ | Work done by gas | J |
| $\Delta U = q + W$ | First law | J |

**Constants:** $R = 8.31\ \mathrm{J\,mol^{-1}K^{-1}}$ $k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ $N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$

## Worked Examples

### Example 1 — Specific Heat Capacity

**Question:** A 2.0 kg block of aluminium ($c = 900\ \mathrm{J\,kg^{-1}K^{-1}}$) is heated from 20 °C to 80 °C. Calculate the energy supplied.

**Solution**

$$Q = mc\Delta T = 2.0 \times 900 \times (80 - 20) = \mathbf{1.08 \times 10^5}\ \mathrm{J}$$

### Example 2 — Ideal Gas Law

**Question:** A fixed mass of ideal gas has pressure $1.2 \times 10^5$ Pa and volume $3.0 \times 10^{-3}$ m$^3$ at 27 °C. It is heated at constant pressure until its volume doubles. Find the final temperature.

**Solution**

$T_1 = 27 + 273.15 = 300.15$ K. At constant pressure, $V \propto T$:

$$T_2 = T_1 \times \frac{V_2}{V_1} = 300.15 \times 2 = \mathbf{600}\ \mathrm{K}\ (\mathbf{327}\ °\mathrm{C})$$

### Example 3 — r.m.s. Speed

**Question:** Calculate the r.m.s. speed of nitrogen molecules ($M = 0.028\ \mathrm{kg\,mol^{-1}}$) at 300 K.

**Solution**

Mass of one molecule: $m = 0.028/6.02 \times 10^{23} = 4.65 \times 10^{-26}$ kg

$$c_{\text{r.m.s.}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 300}{4.65 \times 10^{-26}}} = \sqrt{2.67 \times 10^5} = \mathbf{517}\ \mathrm{m\,s^{-1}}$$

### Example 4 — First Law

**Question:** A gas is compressed. 650 J of work is done on the gas and 200 J of heat flows out. Find the change in internal energy.

**Solution**

$W = +650$ J (work done *on* gas), $q = -200$ J (heat leaves)

$$\Delta U = q + W = -200 + 650 = +\mathbf{450}\ \mathrm{J}$$

Internal energy **increases** by 450 J.

## Practice Exam Questions

## Section A — Short Answer Questions

**Q1.** Define *specific latent heat of vaporisation* and explain why it is greater than the specific latent heat of fusion for the same substance.

*[3 marks]*

**Q2.** State the basic assumptions of the kinetic theory of gases.

*[4 marks]*

**Q3.** Explain what is meant by *internal energy* and state how it differs for an ideal gas compared to a real gas.

*[3 marks]*

**Q4.** A gas at pressure $2.0 \times 10^5$ Pa and temperature 300 K occupies a volume of 0.40 $m^3$. Calculate the number of molecules present.

*[2 marks]*

**Q5.** Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ by combining the kinetic theory equation with the ideal gas equation.

*[3 marks]*

## Section B — Longer Structured Questions

**Q6.** A sample of water of mass 0.50 kg is heated from 20 °C to 100 °C and then completely vaporised.
($c_{\text{water}} = 4200\ \text{J kg}^{-1}\text{K}^{-1}$, $L_v = 2.26 \times 10^6\ \text{J kg}^{-1}$)

(a) Calculate the energy needed to heat the water from 20 °C to 100 °C.

*[2 marks]*

(b) Calculate the energy needed to vaporise the water at 100 °C.

*[2 marks]*

(c) Explain in terms of molecular behaviour why energy is needed to vaporise water even though the temperature does not change.

*[2 marks]*

**Q7.** A fixed mass of ideal gas undergoes the cycle A → B → C → A on a $p$–$V$ diagram where A → B is constant temperature expansion, B → C is constant volume pressure decrease, and C → A is constant pressure compression.

(a) State the change in internal energy during A → B and justify your answer.
*[2 marks]*

(b) Using the first law, find $q$ for A → B given that the work done by the gas is 440 J.
*[2 marks]*

(c) Describe the energy changes during the constant volume process B → C.
*[2 marks]*

## Mark Scheme and Answers

**Q1.** Specific latent heat of vaporisation is the energy per unit mass required to change a substance from liquid to gas at constant temperature [1]; greater than $L_f$ because molecules must be completely separated, requiring work against intermolecular forces over a much greater distance than in melting [2].

**Q2.** Large number of molecules moving randomly in all directions with a range of speeds [1]; occupying negligible volume compared to container [1]; collisions perfectly elastic [1]; intermolecular forces negligible except during collisions; collisions of negligible duration; obey Newton's laws [1].

**Q3.** Internal energy is the sum of random kinetic and potential energies of all molecules [1]; for an ideal gas, $E_p = 0$ so internal energy is kinetic energy only [1]; for a real gas, molecules have potential energy due to intermolecular forces [1].

**Q4.** $N = pV/kT = (2.0 \times 10^5 \times 0.40)/(1.38 \times 10^{-23} \times 300) = \mathbf{1.93 \times 10^{25}}$ molecules [2].

**Q5.** $pV = \frac{1}{3}Nm\langle c^2\rangle$ [1]; equate with $pV = NkT$: $\frac{1}{3}m\langle c^2\rangle = kT$ [1]; so $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ [1].

**Q6(a).** $Q = 0.50 \times 4200 \times 80 = \mathbf{1.68 \times 10^5}$ J [2].

**Q6(b).** $Q = 0.50 \times 2.26 \times 10^6 = \mathbf{1.13 \times 10^6}$ J [2].

**Q6(c).** KE of molecules unchanged (temperature constant) [1]; energy breaks intermolecular bonds as molecules separate completely, increasing potential energy [1].

**Q7(a).** $\Delta U = 0$ [1]; temperature is constant and for an ideal gas internal energy depends only on temperature [1].

**Q7(b).** $\Delta U = 0$, work done on gas $= -440$ J; $q = +440$ J — heat flows into gas [2].

**Q7(c).** No work done ($\Delta V = 0$) [1]; temperature and KE decrease; heat flows out; internal energy decreases [1].

## Revision Checklist

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Explain thermal equilibrium and direction of heat flow | |
| ☐ | Convert between Celsius and Kelvin | |
| ☐ | State examples of thermometric properties | |
| ☐ | Define and use specific heat capacity: $Q = mc\Delta T$ | |
| ☐ | Define and use specific latent heat: $Q = mL$ | |
| ☐ | State the assumptions of kinetic theory | |
| ☐ | Use $pV = nRT$ and $pV = NkT$ | |
| ☐ | Derive and use $pV = \frac{1}{3}Nm\langle c^2\rangle$ | |
| ☐ | Show that mean KE of a molecule is $\frac{3}{2}kT$ | |
| ☐ | Define internal energy and relate to temperature | |
| ☐ | Apply the first law $\Delta U = q + W$ | |
| ☐ | Interpret and use $p$–$V$ diagrams | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

Remember: understanding *why* formulas work is more powerful than memorising them. Practice drawing $p$–$V$ diagrams and deriving key results from first principles.